INDEPENDENT DEMO · SYNTHETIC DATA

Messy spreadsheets.
One clean weekly report.

A reproducible Python workflow that normalizes messy exports, validates every row, logs exceptions, and packages the result as a one-click Excel report.

1,206	1,166	78	0.030s
RAW ROWS	CLEAN ROWS	QUALITY EVENTS	RUNTIME

Built with Python 3.15 + openpyxl · 6/6 automated tests passed · human-reviewed output

01 · PROBLEM

Three exports. Four formats. One weekly headache.

The source files contain mixed dates, duplicate order IDs, inconsistent SKUs, currency symbols, blank prices, invalid discounts, and orphan customer/product references.

Before · raw source rows

Order	Date	SKU	Qty	Price	Discount	Status
ORD-10007	not-a-date	SKU-1001	2	49.29	0.00	cancell
ORD-10019	2026/07/03	SKU-1016	blank	94.86	0.20	paid
ORD-10029	2026-05-15	SKU-1025	3	blank	15%	pendir
ORD-10043	2026-05-10	SKU-1009	3	23.13	blank	refund
ORD-10047	2026-04-12	SKU-1024	-2	23.61	0.00	cancell

After · validated rows

Order	Date	SKU	Qty	Unit price	Disc.	Net revenue	Status
ORD-10001	2026-09-22	SKU-1037	2	142.41	0%	**$284.82**	paid
ORD-10002	2026-07-08	SKU-1017	4	24.44	8%	**$89.94**	paid
ORD-10003	2026-09-17	SKU-1002	2	37.21	8%	**$68.47**	paid
ORD-10004	2026-08-09	SKU-1015	3	46.81	0%	**$0.00**	refunded
ORD-10005	2026-05-17	SKU-1005	1	95.17	0%	**$0.00**	pending

Sample rows shown. The full demo processes 1,206 rows and keeps a row-level exception log.

02 · BUILD

An auditable pipeline, not a black box.

Ingest 1

Reads orders, customers, and products without copy-paste.

Normalize 2

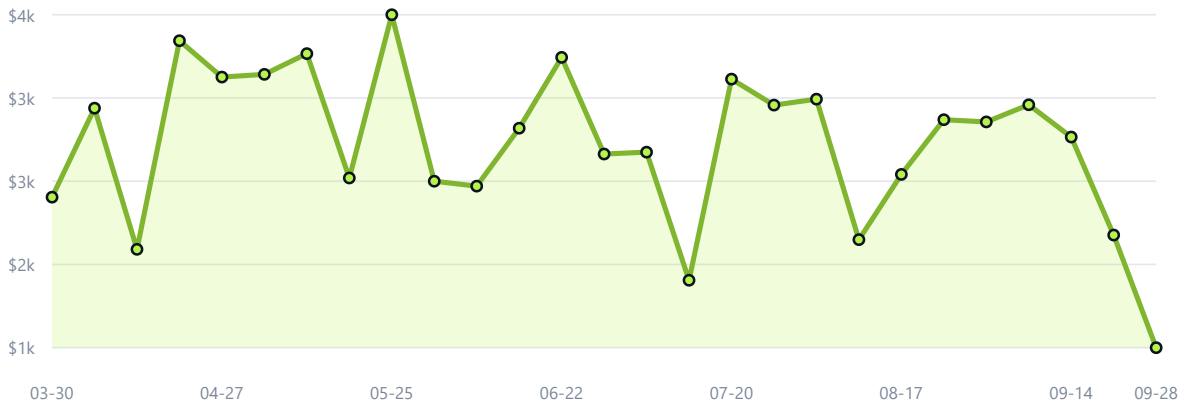
Parses dates, money, SKUs, quantities, discounts, status, and region.

Validate 3

Checks duplicates, referential integrity, numeric bounds, and email format.

Package 4

Writes Excel, CSV, JSON, exception log, and a short walkthrough.



✓ **40 invalid rows excluded**
Every exclusion has a row number and reason.

✓ **6 duplicates removed**
The first occurrence is kept; later rows are logged.

✓ **Paid revenue KPI**
Cancelled/refunded rows are visible but excluded from paid revenue.

03 · DELIVERABLES

What the client receives.

Cleaned workbook

Dashboard, clean orders, exceptions, weekly and category summaries, charts.

Exception log

Row-level audit trail with issue type, severity, and action.

Repeatable script

One command reruns the same logic next week.

Sample outputs

CSV/JSON files to verify before hiring.

README

Install, run, schema, assumptions, and limits.

Walkthrough

Captioned demo video; no voiceover required.

Need this for your files?

Send one sample file and the exact output format. I will confirm fit, scope, and a fixed price before work starts.

Independent demo. Synthetic data only. No client names, logos, testimonials, or revenue claims. AI-assisted build, human-reviewed and tested.